

Normal Probability Function:

A continuous random variable X having the bell-shaped distribution shown in the following figure is called a normal random variable. The mathematical equation of this variable depends on the two parameters μ and σ^2 , known as its mean and variance, respectively. The normal random variable X is frequently denoted by $N(\mu, \sigma^2)$ which is conventionally expressed as $X \sim N(\mu, \sigma^2)$. We call the density function of X , the normal probability density function and denote it by $f(x; \mu, \sigma^2)$.

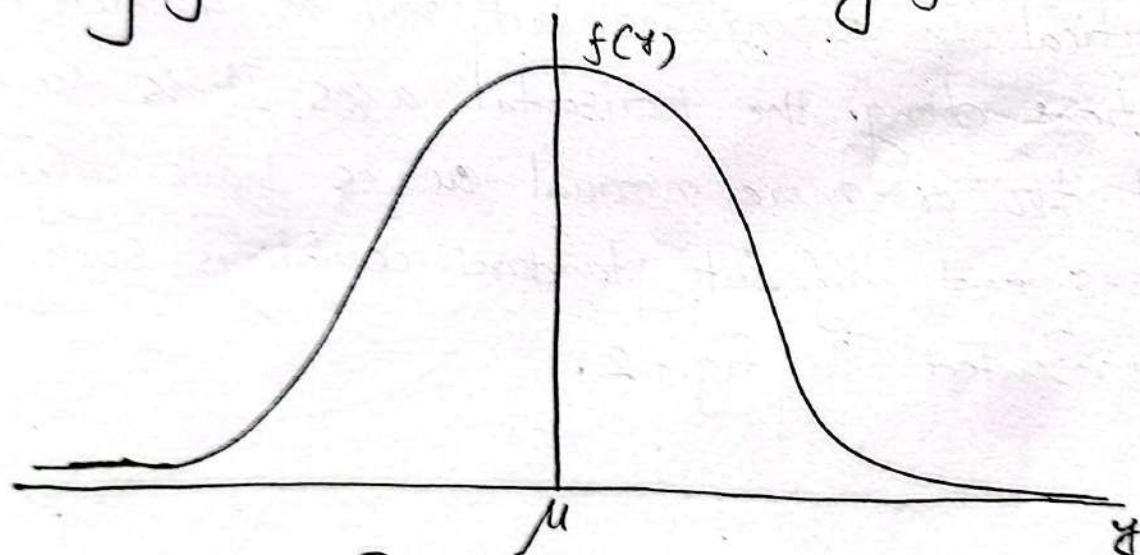


Fig.-1

Defⁿ:

A random variable X is said to have a normal distribution with mean μ and variance σ^2 ($-\infty < \mu < \infty$ and $\sigma^2 > 0$) if X has a continuous distribution for which the probability density function is

$$f(x; \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right], -\infty < x < \infty$$

where $\pi = 3.1416$ and $e = 2.71828$ are two constants

It is important to remember that a normal distribution is completely determined once its mean μ and standard deviation σ are known. This means that once μ and σ are given, then the ordinate $f(x; \mu, \sigma^2)$ can easily be computed for different values of x .

We can easily demonstrate that a normal distribution might have the same standard deviation but different means. In this instance, two or more curves will be identical in appearance but will be centered at different positions along the horizontal axes. It is also possible that two or more normal curves, have different means and different standard deviations. Such a sketch is presented in Fig.-2.

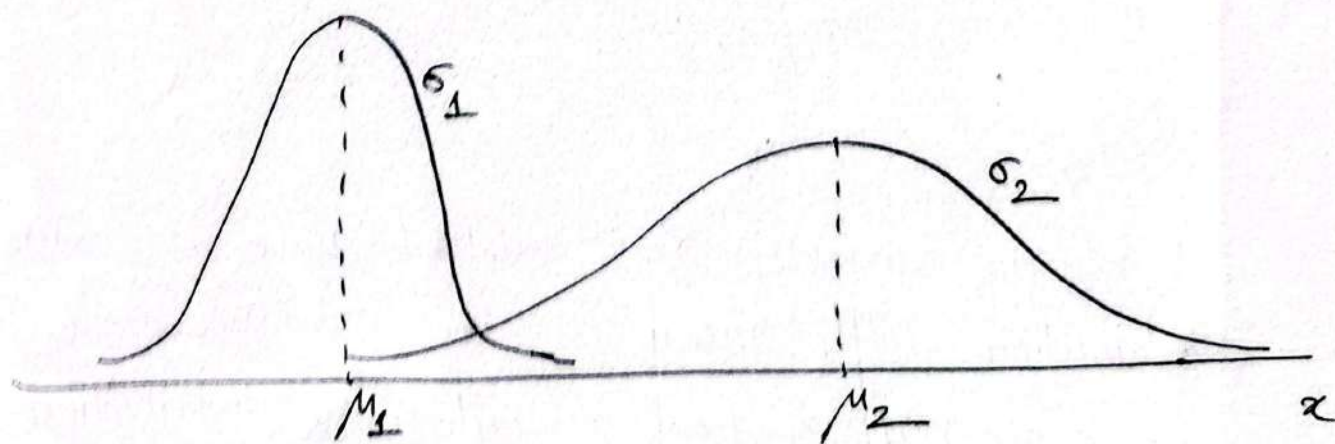


Fig. -2

Finally, we present three more graphs of the normal distribution, which have the same mean but different standard deviations (see Fig. 3). Here the curves are centred at exactly the same position on the horizontal axis. It is important to note that in all cases, the total area under the curves will add to one.

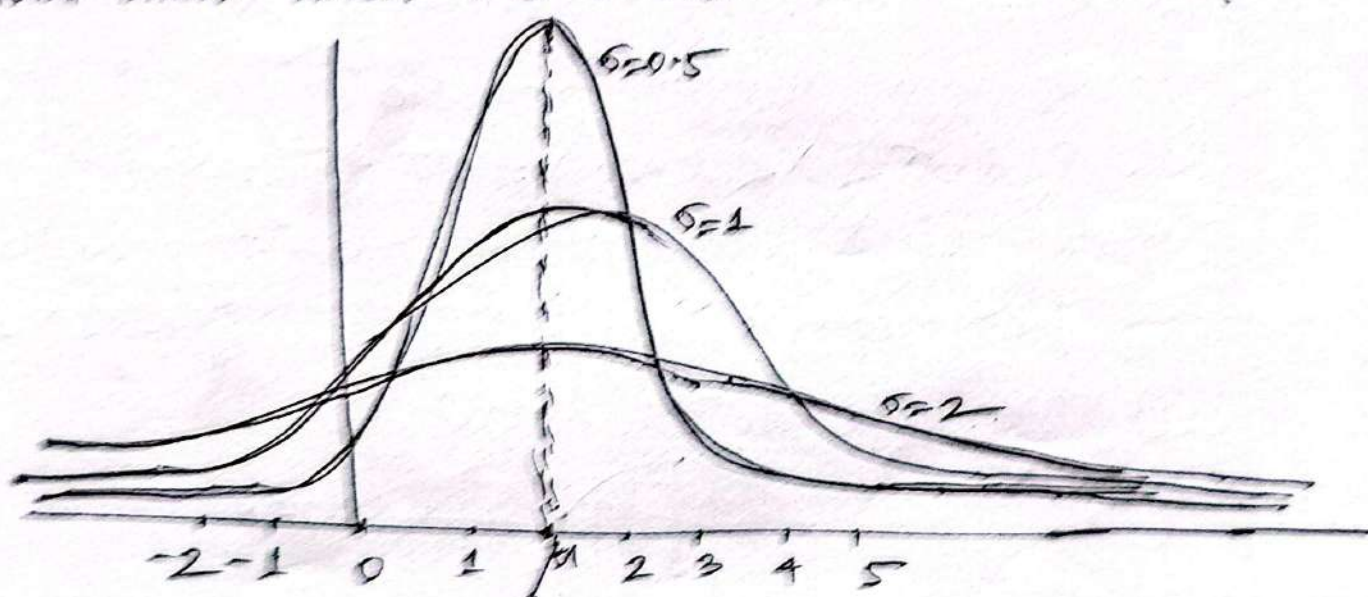
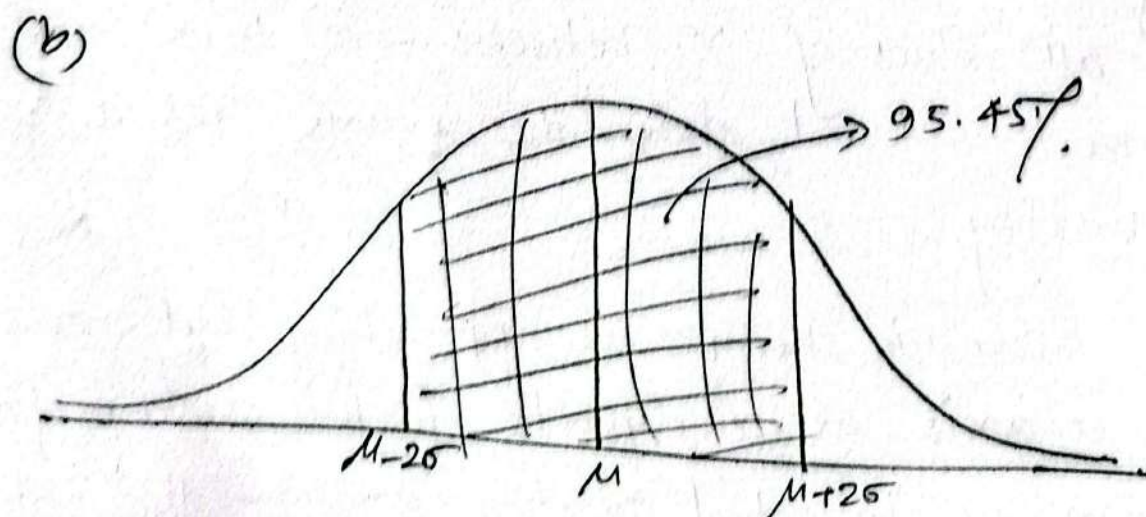
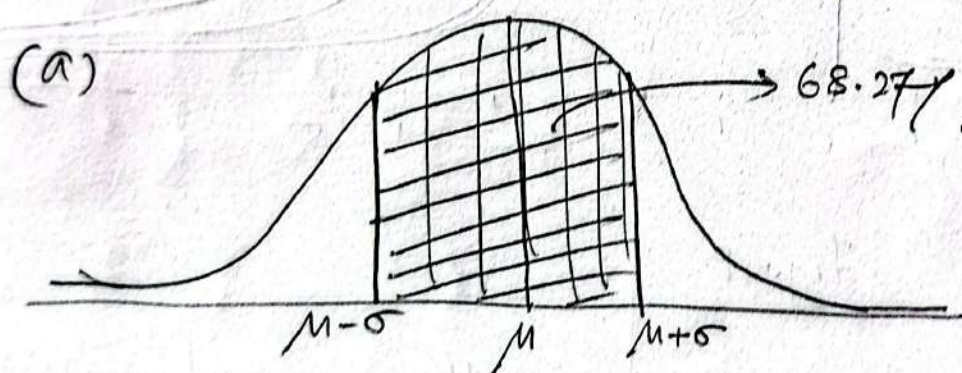


Fig. 3

Theoretically, the domain of the normal variable X is infinite, and the normal distribution is a distribution for all values of X between $-\infty$ to $+\infty$ and each narrow interval along the x -axis has a nonzero probability.

The curve is shown as approaching but never touching the x -axis as x departs further away from the mean μ . In actual practice, however, the value of

$f(x)$, or the ordinate, becomes very close to zero, or almost touches the x -axis, within a short interval. As a matter of fact, about 99.73% of all observations would fall within plus and minus three standard deviation from the mean. In other words, $P(\mu - 3\sigma < X < \mu + 3\sigma) = 99.73\%$. The proportion of X falling within two standard deviation from the mean is 95.45%, i.e., $P(\mu - 2\sigma < X < \mu + 2\sigma) = 95.45\%$. Similarly, the proportion of X falling within one standard deviation from the mean is 68.27%, i.e., $P(\mu - \sigma < X < \mu + \sigma) = 68.27\%$. This area are shown below



(c)

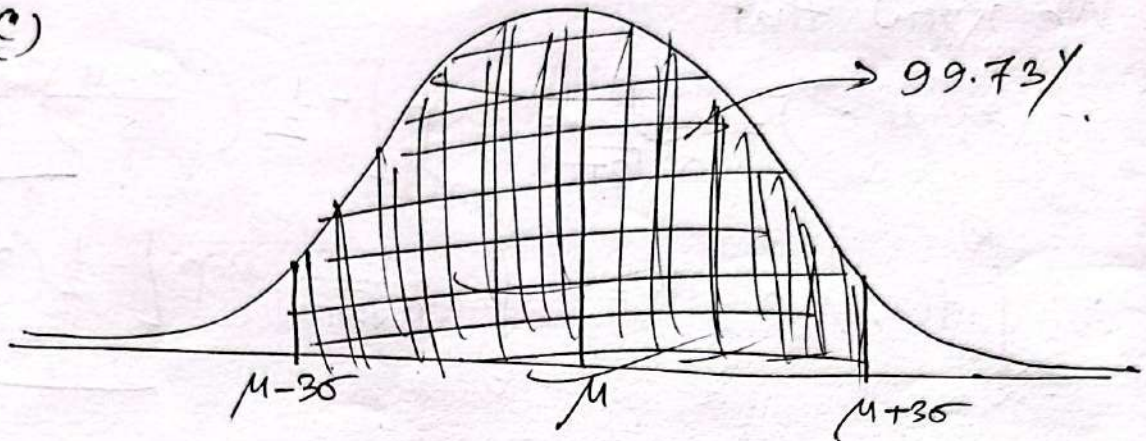


Fig-4 : Area between selected points under the normal curve.

Problem :

Verify that the area under the normal curve having the probability density function defined by the formula

$$f(x; \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2 \right], -\infty < x < \infty$$

is 1.

Proof : Solⁿ :

Here, we need to prove that

$$\int_{-\infty}^{\infty} f(x; \mu, \sigma^2) dx = 1$$

$$\Rightarrow \int_{-\infty}^{\infty} f(x; \mu, \sigma^2) dx = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2 \right] dx = 1$$

We know that

$$f(x; \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$$

$$\therefore \int_{-\infty}^{\infty} f(x; \mu, \sigma^2) dx = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right] dx \quad \text{--- (1)}$$

Let,

$$y = \frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2$$

$$\Rightarrow dy = \frac{1}{2} \cdot 2\left(\frac{x-\mu}{\sigma}\right) \cdot \frac{1}{\sigma} dx$$

$$\Rightarrow dy = \left(\frac{x-\mu}{\sigma}\right) \cdot \frac{1}{\sigma} dx$$

$$\Rightarrow dy = \sqrt{2y} \cdot \frac{1}{\sigma} dx$$

$$\Rightarrow dx = \frac{\sigma}{\sqrt{2y}} dy$$

x	$-\infty$	∞
y	$-\infty$	∞

From (1) we can write,

$$\begin{aligned} \int_{-\infty}^{\infty} f(x; \mu, \sigma^2) dx &= \int_{-\infty}^{\infty} \frac{1}{\cancel{\sigma}\sqrt{2\pi}} \exp[-y] \frac{\cancel{\sigma}}{\sqrt{2y}} dy \\ &= \int_{-\infty}^{\infty} \frac{1}{2\sqrt{\pi}} y^{-1/2} e^{-y} dy \end{aligned}$$

$$\begin{aligned}
 &= \frac{2}{2\sqrt{\pi}} \int_0^{\infty} y^{-1/2} e^{-y} dy \\
 &= \frac{1}{\sqrt{\pi}} \int_0^{\infty} y^{\frac{1}{2}-1} e^{-y} dy \\
 &= \frac{1}{\sqrt{\pi}} \Gamma\left(\frac{1}{2}\right) \quad \left[\because \Gamma(n) = \int_0^{\infty} y^{n-1} e^{-y} dy \right] \\
 &= \frac{1}{\sqrt{\pi}} \cdot \sqrt{\pi} \\
 &= 1.
 \end{aligned}$$

$$\therefore \int_{-\infty}^{\infty} f(x; \mu, \sigma^2) dx = 1.$$

Proved,

Problem:

Find the mean and variance of the normal distribution.

Proof:

We know that the density function of a random variable X following a normal distribution is given by

$$f(x, \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp \left[-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2 \right]$$

Now,

$$\text{Mean of } X = E[X] = \int_{-\infty}^{\infty} x f(x; \mu, \sigma^2) dx$$

$$\Rightarrow E[X] = \int_{-\infty}^{\infty} x \cdot \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right] dx$$

$$\therefore E[X] = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} x \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right] dx$$

①

let,

$$y = \frac{x-\mu}{\sigma}$$

$$\Rightarrow x = \mu + \sigma y$$

$$\therefore dx = \sigma dy$$

L.C :

x	$-\infty$	∞
y	$-\infty$	∞

① $\Rightarrow$

$$E[X] = \frac{1}{\cancel{\sigma}\sqrt{2\pi}} \int_{-\infty}^{\infty} (\mu + \cancel{\sigma}y) e^{-\frac{1}{2}y^2} \cancel{\sigma} dy$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (\mu + \sigma y) e^{-\frac{1}{2}y^2} dy$$

$$= \frac{\mu}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}y^2} dy + \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y e^{-\frac{1}{2}y^2} dy$$

$$= \frac{\mu}{\sqrt{2\pi}} \cdot 2 \int_0^{\infty} e^{-\frac{1}{2}y^2} dy + 0 \quad \left[\begin{array}{l} \text{even function} \\ \text{odd function} \end{array} \right]$$

$$\Rightarrow E[X] = \mu \cdot \frac{\sqrt{2}}{\sqrt{\pi}} \int_0^{\infty} e^{-\frac{1}{2}t^2} dy \quad \text{--- (2)}$$

Let, $\frac{y^2}{2} = z$

$$\frac{2y}{2} \cdot dy = dz$$

$$\Rightarrow y \, dy = dz$$

$$\Rightarrow dy = \frac{1}{y} dz = \frac{1}{\sqrt{2z}} dz$$

y	0	∞
z	0	∞

(2) $\Rightarrow$

$$E[X] = \mu \cdot \frac{\sqrt{2}}{\sqrt{\pi}} \int_0^{\infty} e^{-z} \cdot \frac{1}{\sqrt{2z}} dz$$

$$= \mu \cdot \frac{1}{\sqrt{\pi}} \int_0^{\infty} z^{-\frac{1}{2}} e^{-z} dz$$

$$= \mu \cdot \frac{1}{\sqrt{\pi}} \int_0^{\infty} z^{\frac{1}{2}-1} e^{-z} dz$$

$$= \mu \cdot \frac{1}{\sqrt{\pi}} \cdot \Gamma\left(\frac{1}{2}\right) = \mu \cdot \frac{1}{\sqrt{\pi}} \cdot \sqrt{\pi}$$

$$\therefore E[X] = \mu$$

$\therefore$ Mean of normal distribution
 $E[X] = \mu$

Again,

$$V[X] = E[(X - \mu)^2]$$

$$= \int_{-\infty}^{\infty} (x - \mu)^2 f(x; \mu, \sigma^2) dx$$

$$= \int_{-\infty}^{\infty} (x - \mu)^2 \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x - \mu}{\sigma}\right)^2\right] dx$$

$$\Rightarrow V[X] = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (x - \mu)^2 \exp\left[-\frac{1}{2}\left(\frac{x - \mu}{\sigma}\right)^2\right] dx \quad \text{--- (3)}$$

Let,

$$y = \frac{x - \mu}{\sigma}$$

$$\Rightarrow x = \mu + \sigma y$$

$$\therefore dx = \sigma dy$$

x	$-\infty$	∞
y	$-\infty$	∞

$$\therefore V[X] = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \sigma^2 y^2 \exp\left[-\frac{1}{2} y^2\right] \sigma dy$$

$$\Rightarrow V[X] = \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y^2 e^{-\frac{1}{2} y^2} dy \quad \text{--- (4)}$$

Again let,

$$z = \frac{1}{2} y^2$$

$$\Rightarrow dz = \frac{1}{2} 2y dy \Rightarrow dz = y dy$$

$$\Rightarrow dy = \frac{1}{y} dz = \frac{1}{\sqrt{2z}} dz$$

1	$-\infty$	∞
2	$-\infty$	∞

④ $\Rightarrow$

$$V[X] = \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} 2z \cdot e^{-z} \cdot \frac{1}{\sqrt{2z}} dz$$

$$= \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \sqrt{2z} e^{-z} dz$$

$$= \frac{\sigma^2}{\sqrt{\pi}} \cdot 2 \int_0^{\infty} z^{\frac{1}{2}} e^{-z} dz$$

$$= \frac{2\sigma^2}{\sqrt{\pi}} \int_0^{\infty} z^{\frac{3}{2}-1} e^{-z} dz$$

$$= \frac{2\sigma^2}{\sqrt{\pi}} \Gamma\left(\frac{3}{2}\right) = \frac{2\sigma^2}{\sqrt{\pi}} \Gamma\left(\frac{1}{2}+1\right)$$

$$= \frac{2\sigma^2}{\sqrt{\pi}} \cdot \frac{1}{2} \cdot \Gamma\left(\frac{1}{2}\right) \quad [\because \Gamma(n+1) = n \Gamma(n)]$$

$$= \frac{\sigma^2}{\sqrt{\pi}} \Gamma\left(\frac{1}{2}\right) = \frac{\sigma^2}{\sqrt{\pi}} \cdot \sqrt{\pi} = \sigma^2$$

$\therefore$ Variance of Normal distribution: $V[X] = \sigma^2$

Standard normal distribution:

If a random variable X has a normal distribution with mean μ and variance σ^2 (i.e., $X \sim N(\mu, \sigma^2)$) then the variable $Z = \frac{X - \mu}{\sigma}$, will be called a standard normal variate (or Z-score) and its distribution is referred to as the standard normal distribution having the following density function:

$$f(z; 0, 1) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty$$

We verify below that

$$E[Z] = 0 \quad \& \quad V[Z] = 1$$

Since, $Z = \frac{X - \mu}{\sigma}$

$$\begin{aligned} \therefore E[Z] &= E\left[\frac{X - \mu}{\sigma}\right] \\ &= E\left[\frac{X}{\sigma}\right] - E\left[\frac{\mu}{\sigma}\right] \\ &= \frac{1}{\sigma} E[X] - \frac{\mu}{\sigma} E[1] \\ &= \frac{1}{\sigma} \cdot \mu - \frac{\mu}{\sigma} \cdot 1 \\ &= \frac{\mu}{\sigma} - \frac{\mu}{\sigma} = 0 \\ \therefore E[Z] &= 0 \end{aligned}$$

$$\therefore V[Z] = E[Z^2] - \{E[Z]\}^2$$

$$= E\left[\left(\frac{X-\mu}{\sigma}\right)^2\right] - (0)^2$$

$$= \frac{1}{\sigma^2} E[(X-\mu)^2] - 0$$

$$= \frac{1}{\sigma^2} E[X^2 - 2X\mu + \mu^2]$$

$$= \frac{1}{\sigma^2} \{E[X^2] - 2\mu E[X] + \mu^2 E[1]\}$$

$$= \frac{1}{\sigma^2} \{E[X^2] - 2\mu \cdot \mu + \mu^2\}$$

$$= \frac{1}{\sigma^2} \{E[X^2] - 2\mu^2 + \mu^2\}$$

$$\therefore V[Z] = \frac{1}{\sigma^2} \{E[X^2] - \mu^2\} \quad \text{--- ①}$$

$$\text{Now, } E[X^2] = \int_{-\infty}^{\infty} x^2 f(x; \mu, \sigma^2) dx$$

$$= \int_{-\infty}^{\infty} x^2 \cdot \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right] dx$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} x^2 \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right] dx$$

$$\textcircled{1} \Rightarrow V[Z] = \frac{1}{\sigma^2} \left\{ \sigma^2 + \mu^2 - \mu^2 \right\}$$

$$\therefore V[Z] = 1$$

Thus $Z \sim N(0, 1)$

The cumulative distribution function (CDF) of the standard normal variate Z is usually denoted by the symbol $\phi(z)$. Thus,

$$\phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-z^2/2} dz$$

The value of $\phi(z)$ given by the above ^{definite} integral is called the probability integral or the error function.



Since the pdf of the standard normal distribution is symmetric with respect to the point $x=0$, it follows that $P(X \leq x) = P(X \geq -x)$ for any number x ($-\infty < x < \infty$).

Since $P(X \leq x) = \phi(x)$ and $P(X \geq -x) = 1 - \phi(-x)$, we have $\phi(x) + \phi(-x) = 1$ for $-\infty < x < \infty$

Problem:

In a normal distribution with mean 50 and variance 100, what are the values that cover central 50% of the area under the curve?

Solⁿ:

From the Normal table, we look for the values of K such that

$$P(-K_1 < Z < K_2) = 0.50$$

Because of symmetry, we can take $K_1 = K_2 = K$, therefore

$$P(-K_1 < Z < K_2) = 2 P(Z < K) = 0.50$$

Thus, $P(Z < K) = 0.25$

From the table, we find that $P(Z < -0.675) = 0.25$

Hence,

$$P(-0.675 < Z < 0.675) = 0.50$$

Therefore, we can write,

$$\frac{x_1 - 50}{10} = -0.675 \quad \& \quad \frac{x_2 - 50}{10} = 0.675$$

$$\Rightarrow x_1 = 43.25$$

$$\Rightarrow x_2 = 56.75$$

Therefore, $P(43.25 < X < 56.75) = 0.50$

Problem:

The normal random variable X having a mean 30 and unknown variance is so distributed that $P(X < 35) = 0.69$. Find the variance of X .

Solⁿ:

Let σ^2 be the variance of the normal R.V X .

Then,

$$P(X < 35) = P\left(Z < \frac{35-30}{\sigma}\right) = 0.69$$

From the normal table, it is found that

$$P(Z < 0.50) = 0.69$$

Therefore,

$$\frac{35-30}{\sigma} = 0.50$$

$$\Rightarrow \sigma = 10$$

$$\Rightarrow \sigma^2 = 100$$

Ans)